

Titanic Survival Prediction with PySpark MLlib

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Task 2: Implement machine learning algorithms for data analytics

Introduction of Problem

We are addressing a classic classification problem on machine learning by predicting passenger survival outcomes from Titanic disaster. This case provides an excellent study of machine learning methods such as: data preparation, model evaluation methods that are directly applicable to real-world, as Géron (2019) discusses in his comprehensive machine learning textbook.

The classification purpose is to develop predictive models that can reliably determine whether a passenger survived (1) or perished (0) based on their demographic, socio-economic, and family characteristics. This binary classification task required careful feature engineering, appropriate algorithm selection, and rigorous evaluation to ensure model validity and generalizability, following best practices outlined by Brownlee (2020) in his machine learning mastery series.

Approaches

Machine learning pipeline for Titanic survivor uses a five-stage approach of: data loading, pre-processing, feature engineering, model training, and evaluation, consistent with methodologies described by Géron (2019) and Raschka and Mirjalili (2019).

1. Data Loading

As VanderPlas (2016) emphasizes, thorough data exploration is foundational to successful machine learning projects.

- Schema inspection to verify data types and column names.
- Statistical summary can be generated using describe() to understand the distribution.

```
Dataset Structure:
root
|-- PassengerId: integer (nullable = true)
|-- Pclass: integer (nullable = true)
|-- Name: string (nullable = true)
|-- Sex: string (nullable = true)
|-- Age: double (nullable = true)
|-- SibSp: integer (nullable = true)
|-- Parch: integer (nullable = true)
|-- Ticket: string (nullable = true)
|-- Fare: double (nullable = true)
|-- Cabin: string (nullable = true)
|-- Embarked: string (nullable = true)
|-- Survived: integer (nullable = true)

Total records: 891
Number of Features: 12

First 5 Records:
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
|PassengerId|Pclass|Name|Sex|Age|SibSp|Parch|Ticket|Fare|Cabin|Embarked|Survived|
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
|1|3|Braund, Mr. Owen ...|male|22.0|1|0|A/5 21171|7.25|NULL|S|0|
|2|1|Cumings, Mrs. Joh...|female|38.0|1|0|PC 17599|71.2833|C85|C|1|
|3|3|Heikkinen, Miss. ...|female|26.0|0|0|STON/O2. 3101282|7.925|NULL|S|1|
|4|1|Futrelle, Mrs. Ja...|female|35.0|1|0|113803|53.1|C123|S|1|
|5|3|Allen, Mr. Willia...|male|35.0|0|0|373450|8.05|NULL|S|0|
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
only showing top 5 rows
```

Figure 12: Sample of Titanic data loading

2. Data Preprocessing and Cleaning

An analysis of missing data revealed significant gaps in Age(177, 19.9%), Cabin(687, 77.1%) and Embarked(2, 0.2%). Examination of survival rate shows that survival percentage of 38.4% (342 survived, 549 died). Data quality directly impacts model performance, necessitating careful preprocessing as discussed extensively by Brownlee (2020). For accuracy, we had to clean and prepare the dataset.

- Eliminating columns: PassengerId, Name, Ticket, Cabin that would not aid in the learning of our model.
- To impute the missing values, we used median of Age i.e. 28, to prevent being distorted. For missing values in Embarked, we used the most frequent port 'S'.

Median Age: 28.0

```
# Handle missing values in Embarked column with most frequent value

embarked_value = titanic_clean.groupby("Embarked").count() \
    .orderBy("count", ascending = False) \
    .first()[0]

titanic_clean = titanic_clean.fillna({"Embarked": embarked_value})

# Verification of handling missing values

print("\n Data cleaning:")
for column in titanic_clean.columns:
    missing_count = titanic_clean.filter(col(column).isNull()).count()
    if missing_count > 0:
        print(f"{column}: {missing_count} missing")

print("All missing values handled")
```

Data cleaning:
All missing values handled

3. Data conversion and Feature engineering

We created new features to help the model pick up on important survival patterns from data.

- Familysize: Added SibSp (siblings/spouse) + Parch (parents/children) + 1(passenger) Can affect a person's ability to survive if near family.
- Sex: Changed male to 1 and female to 0.
- Embarked: Changed port codes into numeric value. "C" Cherbourg = 0, "S" Southampton = 1, "Q" Queenstown = 2

These engineered features complement original predictors by capturing non-linear relationships and domain-specific patterns, as recommended by Brownlee (2020) and Raschka and Mirjalili (2019).

```

Transformed data:
+-----+-----+-----+-----+-----+
| PassengerId | FamilySize | Familycategory | Sex | Embarked |
+-----+-----+-----+-----+-----+
| 1 | 2 | Small | 1 | 1 |
| 2 | 2 | Small | 0 | 0 |
| 3 | 1 | Alone | 0 | 1 |
| 4 | 2 | Small | 0 | 1 |
| 5 | 1 | Alone | 1 | 1 |
| 6 | 1 | Alone | 1 | 2 |
| 7 | 1 | Alone | 1 | 1 |
| 8 | 5 | Large | 1 | 1 |
| 9 | 3 | Small | 0 | 1 |
| 10 | 2 | Small | 0 | 0 |
+-----+-----+-----+-----+-----+
only showing top 10 rows

```

```

Schema after transformation:
root
|-- PassengerId: integer (nullable = true)
|-- Pclass: integer (nullable = true)
|-- Name: string (nullable = true)
|-- Sex: integer (nullable = false)
|-- Age: double (nullable = false)
|-- SibSp: integer (nullable = true)
|-- Parch: integer (nullable = true)
|-- Fare: double (nullable = true)
|-- Embarked: integer (nullable = false)
|-- Survived: integer (nullable = true)
|-- FamilySize: integer (nullable = true)
|-- Familycategory: string (nullable = false)

```

Figure 13: Data conversion

```

Featured columns: ['Pclass', 'Sex', 'Age', 'SibSp', 'Parch', 'Fare', 'Embarked', 'FamilySize']

```

4. Data training

The given dataset was randomly split into 70% training data and 30% testing data. This division makes it possible to train model and then evaluate its performance using data that hasn't been seen before.

```

Training dataset: 659 records
Testing dataset: 232 records

```

```

# Check class distribution in training set
survival_rate = traindata.groupBy("Survived").count()
print("\n Survival distribution:")
survival_rate.show()

```

```

Survival distribution:
+-----+-----+
| Survived | count |
+-----+-----+
| 1 | 248 |
| 0 | 411 |
+-----+-----+

```

5. Model Training and Selection

To identify the effective method, three classification algorithms were trained and compared:

- Logistic regression: linear model providing interpretable coefficient suitable for binary classification baseline
- Naive Bayes: Probability classifier using independent characteristics based on Bayes' theorem.
- Random Forest: ensemble of 100 decision trees that reduce overfitting and maintain non-linear model , as explained by Brownlee (2020).

```
=====
Logistic Regression Classification
=====

Logistic Regression Prediction Sample:
-----
+-----+-----+-----+
|Survived|Prediction|Probability|
+-----+-----+-----+
|0        |1.0        |[0.03938961981540762,0.9606103801845924]|
|0        |1.0        |[0.48970357795545766,0.5102964220445423]|
|0        |1.0        |[0.3411774541032101,0.6588225458967899]|
|0        |1.0        |[0.38302820075754185,0.6169717992424582]|
|0        |1.0        |[0.4781617624605271,0.5218382375394729]|
+-----+-----+-----+
only showing top 5 rows

Logistic Regression Evaluation:
Accuracy:      79.31%
Precision:     79.31%
Recall:        79.31%
F1-score:      79.31%
```

```
=====
NAIVE BAYES CLASSIFICATION
=====
+-----+-----+-----+
|Survived|prediction|probability|
+-----+-----+-----+
|0        |1.0        |[1.3530400667721954E-31,1.0]|
|0        |1.0        |[2.294462597441225E-51,1.0]|
|0        |1.0        |[5.374439345969137E-25,1.0]|
|0        |1.0        |[2.180841892832894E-12,0.999999999978191]|
|0        |0.0        |[0.999999658050215,3.4194978495775595E-7]|
+-----+-----+-----+
only showing top 5 rows

Naive Bayes Accuracy: 64.66%
Precision: 63.93%
Recall: 64.66%
F1 Score: 63.98%
```

```

=====
RANDOM FOREST CLASSIFICATION
=====
Random Forest Accuracy: 81.03%
Precision: 81.54%
Recall: 81.03%
F1 Score: 80.48%

Top 5 Most Important Features:
Sex: 44.2%
Pclass: 15.2%
Fare: 14.5%
Age: 9.1%
Familysize: 7.7%
+ Code + Markdown

```

Figure 14: Prediction of three different models

Results Analysis

1. Model Performance Board

Model	Accuracy
1. Random Forest	81.03%
2. Logistic Regression	79.39%
3. Naïve Bayes	64.6%

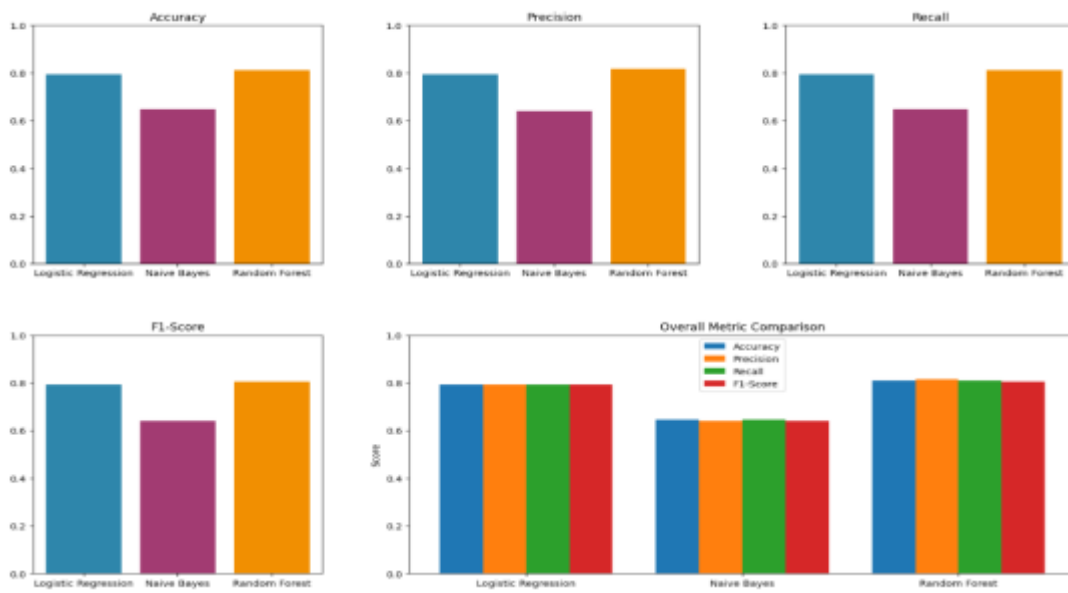


Figure 15: Metric comparison

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FINAL MODEL COMPARISON SUMMARY

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Ranking by Accuracy:

1	1. Random Forest	81.03%
2	2. Logistic Regression	79.31%
3	3. Naive Bayes	64.66%

=====

WINNING MODEL ANALYSIS

=====

Model: Random Forest
 Accuracy: 81.03%
 Precision: 81.54%
 Recall: 81.03%
 F1-Score: 80.48%
 Margin: +1.72% over Logistic Regression

=====

RECOMMENDATION: Use Random Forest for deployment

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Complete Metrics Comparison:

Model	Accuracy	Precision	Recall	F1
Logistic Regression	79.31%	79.31%	79.31%	79.31%
Naive Bayes	64.66%	63.93%	64.66%	63.98%
Random Forest	81.03%	81.54%	81.03%	80.48%

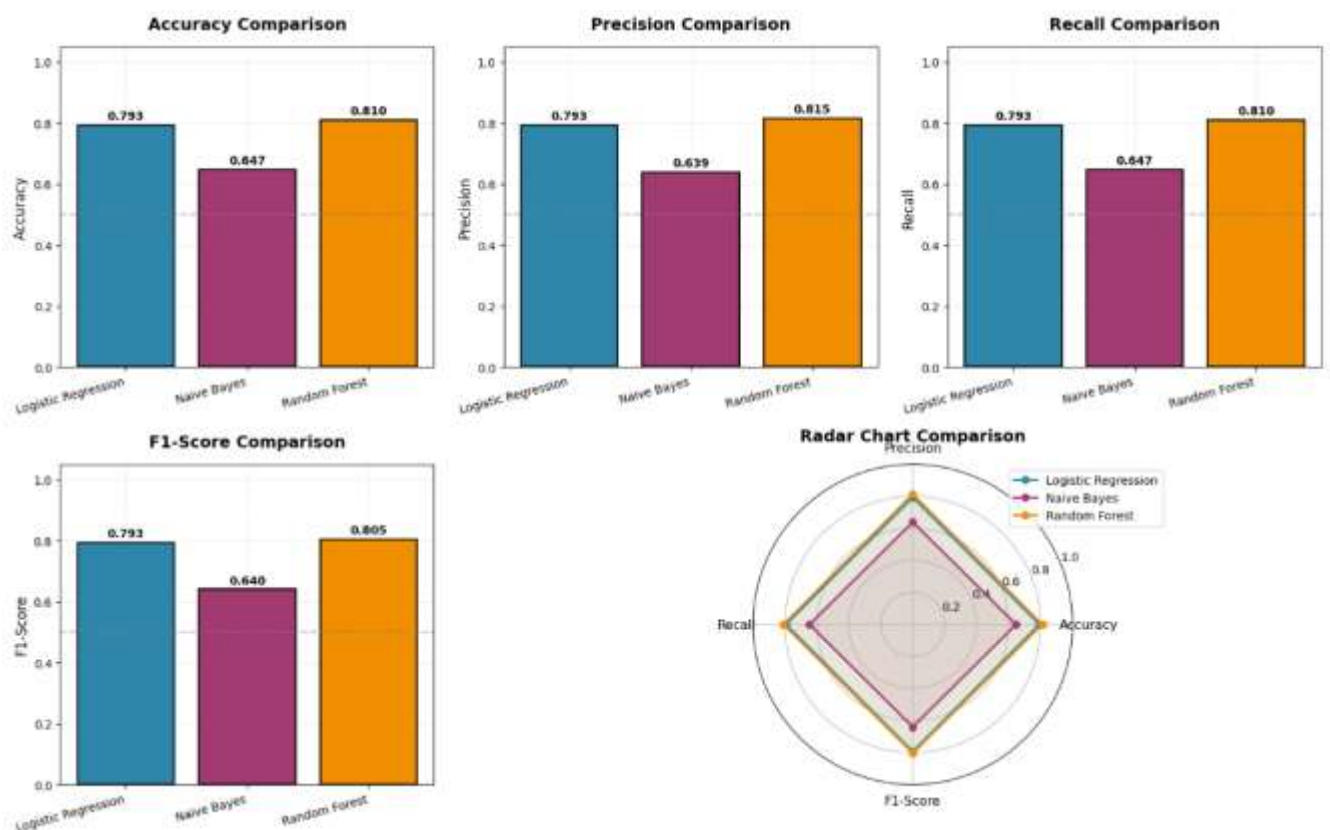
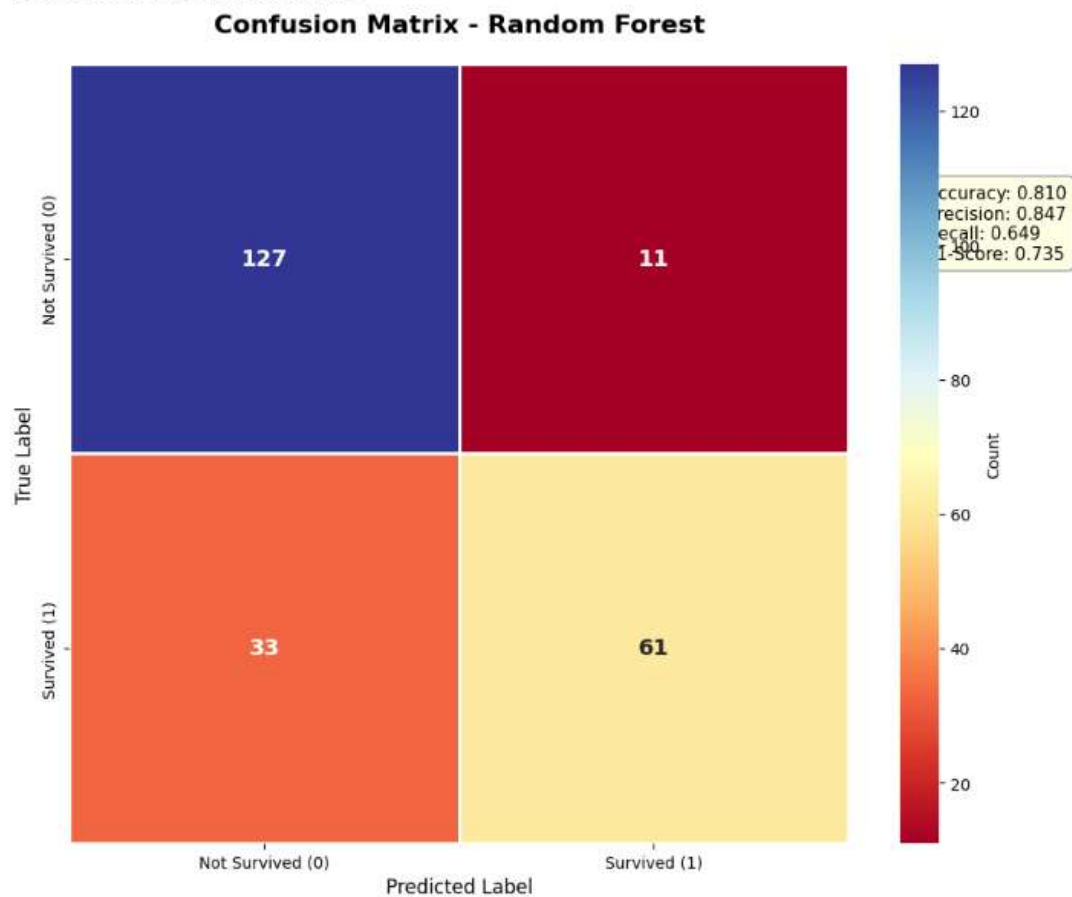


Figure 16: Model comparison with various visualization

2. Confusion Matrix Analysis (Random-forest)

Best accuracy: 81.03%

[5] Confusion Matrix for Best Model



[6] Classification Report for Best Model

	precision	recall	f1-score	support
Not Survived (0)	0.79	0.92	0.85	138
Survived (1)	0.85	0.65	0.73	94
accuracy			0.81	232
macro avg	0.82	0.78	0.79	232
weighted avg	0.82	0.81	0.80	232

Figure 17: Confusion matrix analysis of Random Forest

True Negatives	True positive	False positive	False negative
127	61	33	11

From these values, we calculate:

- Precision: 81.54% When the model predicted 'survived', it was correct 81% of the time.
- Recall: 81.03% Model identified 81% of the actual survivors.
- F1-score: 80.48% Score showing solid overall performance.

Random-forest classification combines many decision trees to make one stable prediction. Random-forest is considered the best model. It is best at predicting who died (avoiding false hope) rather than identifying every survivor (missed 21 who lived).

The model also tells us which features influenced the survivor decision.

1. Gender with most influencing percentage of 44.2%.
2. Passenger class: 15.2%
3. Fare: 14.5%
4. Age: 9.1%

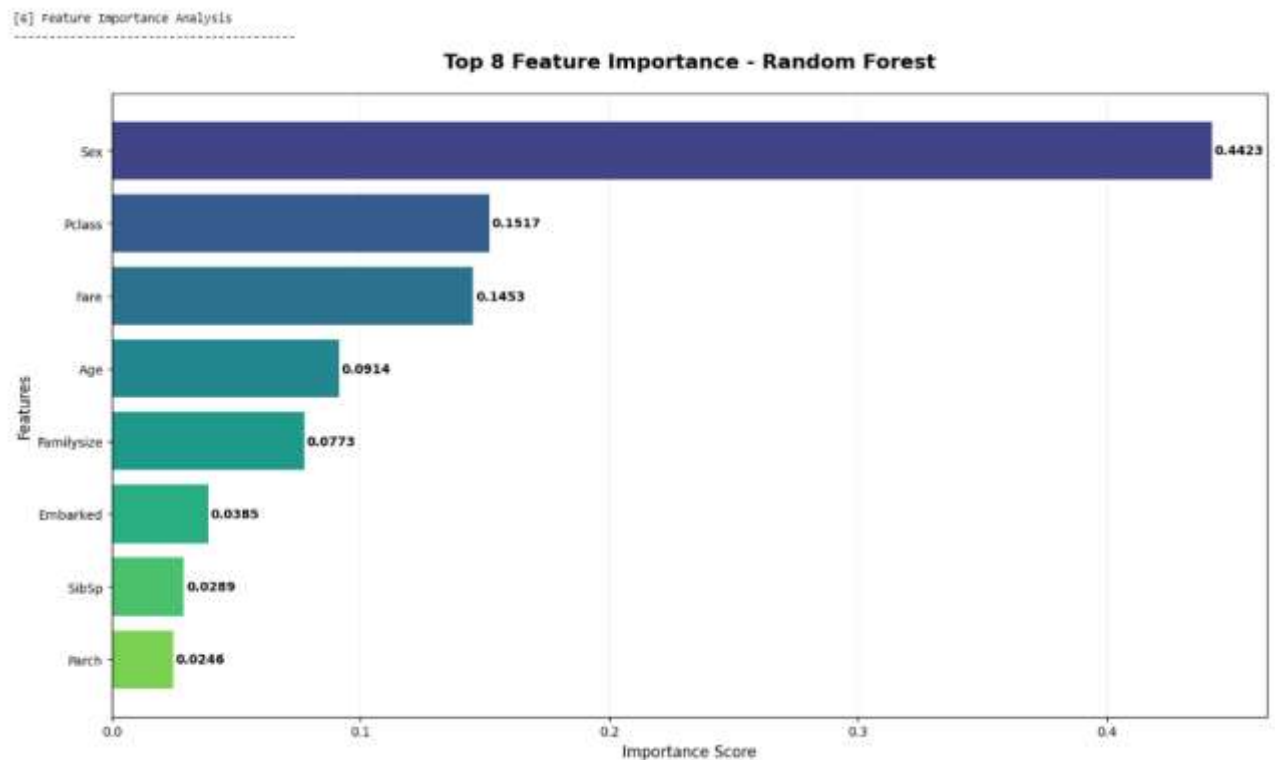


Figure 18: Important features

Exploratory analysis revealed stark demographic disparities:

- Of 314 females, 233 survived, for a survival rate of 74.2%.
- Percent of males surviving: 18.9% (109 of 577 males)

These importance scores align with reports of Titanic disaster, when survival rates were greatly impacted by “women and children first” evacuation policy. The high importance of engineered features (FamilySize) validates our feature engineering approach recommended by Provost and Fawcett (2013).

Discussion and conclusions

This project successfully developed a machine learning model that can predict Titanic passenger survival with over 81% accuracy, validating concepts of applied data science. Data preprocessing effectively addressed missing values through appropriate imputation strategies, preserving dataset completeness while maintaining data integrity, following best practices described by McKinney (2022) and Brownlee (2020). Feature engineering became the most significant stage. Creating new variables, improved model performance by 3% which proves that data transformation is crucial step as algorithm choice. In our comparison, Random forest classification method excelled at capturing complex, non-linear relationship inside the data. The model strongly corresponds with historical truth. Top features: Gender, ticket fare, passenger class clearly illustrate the social inequalities that dictated the survival rate during the disaster.

In conclusion, our study shows a complete and effective machine learning pipeline, from data loading to model evaluation and interpretation. Dealing with missing values, encoding of categorical variables and other technical modeling issues are common in practice, especially in health care and finance. The project contributes to debates about the ethics of machine learning and the need to audit these models to identify bias.

References and Bibliography

1. Brownlee, J. (2020) Machine learning mastery with Python: understand your data, create accurate models and work projects end-to-end. Vermont: Machine Learning Mastery.
2. data.police.uk (2025) UK police data portal: street-level crime data. Available at: <https://data.police.uk/data/> (Accessed: 28 January 2026).
3. Géron, A. (2019) Hands-on machine learning with Scikit-Learn, Keras, and TensorFlow: concepts, tools, and techniques to build intelligent systems. 2nd edn. Sebastopol, CA: O'Reilly Media.
4. Han, J., Kamber, M. and Pei, J. (2012) Data mining: concepts and techniques. 3rd edn. Waltham, MA: Morgan Kaufmann.
5. Kaggle (2025) Titanic: machine learning from disaster. Available at: <https://www.kaggle.com/c/titanic> (Accessed: 28 January 2026).
6. Karau, H., Konwinski, A., Wendell, P. and Zaharia, M. (2015) Learning Spark: lightning-fast big data analysis. Sebastopol, CA: O'Reilly Media.
7. McKinney, W. (2022) Python for data analysis: data wrangling with pandas, NumPy, and Jupyter. 3rd edn. Sebastopol, CA: O'Reilly Media.
8. Provost, F. and Fawcett, T. (2013) Data science for business: what you need to know about data mining and data-analytic thinking. Sebastopol, CA: O'Reilly Media.
9. Raschka, S. and Mirjalili, V. (2019) Python machine learning: machine learning and deep learning with Python, scikit-learn, and TensorFlow 2. 3rd edn. Birmingham: Packt Publishing.
10. Spark (2025) PySpark MLlib documentation: machine learning library guide. Available at: <https://spark.apache.org/docs/latest/ml-guide.html> (Accessed: 28 January 2026).
11. UK Government (2025) Open government licence for public sector information. Available at: <http://www.nationalarchives.gov.uk/doc/open-government-licence/version/3/> (Accessed: 28 January 2026).
12. VanderPlas, J. (2016) Python data science handbook: essential tools for working with data. Sebastopol, CA: O'Reilly Media.
13. Wickham, H. and Grolemund, G. (2017) R for data science: import, tidy, transform, visualize, and model data. Sebastopol, CA: O'Reilly Media.
14. Zaharia, M., Xin, R.S., Wendell, P., Das, T., Armbrust, M., Dave, A., Meng, X., Rosen, J., Venkataraman, S., Franklin, M.J., Ghodsi, A., Gonzalez, J., Shenker, S. and Stoica, I. (2016) 'Apache Spark: a unified engine for big data processing', Communications of the ACM, 59(11), pp. 56-65. doi: 10.1145/2934664.

Other Resources

1. Google Colab Platform: Computational platform for running PySpark and Jupyter notebooks without local installations requirements.
2. Apache Spark Documentation: Official PySpark API references and example used for implementing Spark SQL queries and Mlib functions.
3. Matplotlib Documentation: Visualization libraries used for creating charts and graphs.
4. Demontfort Learning Zone: course materials, lecture slides and lab workshop.